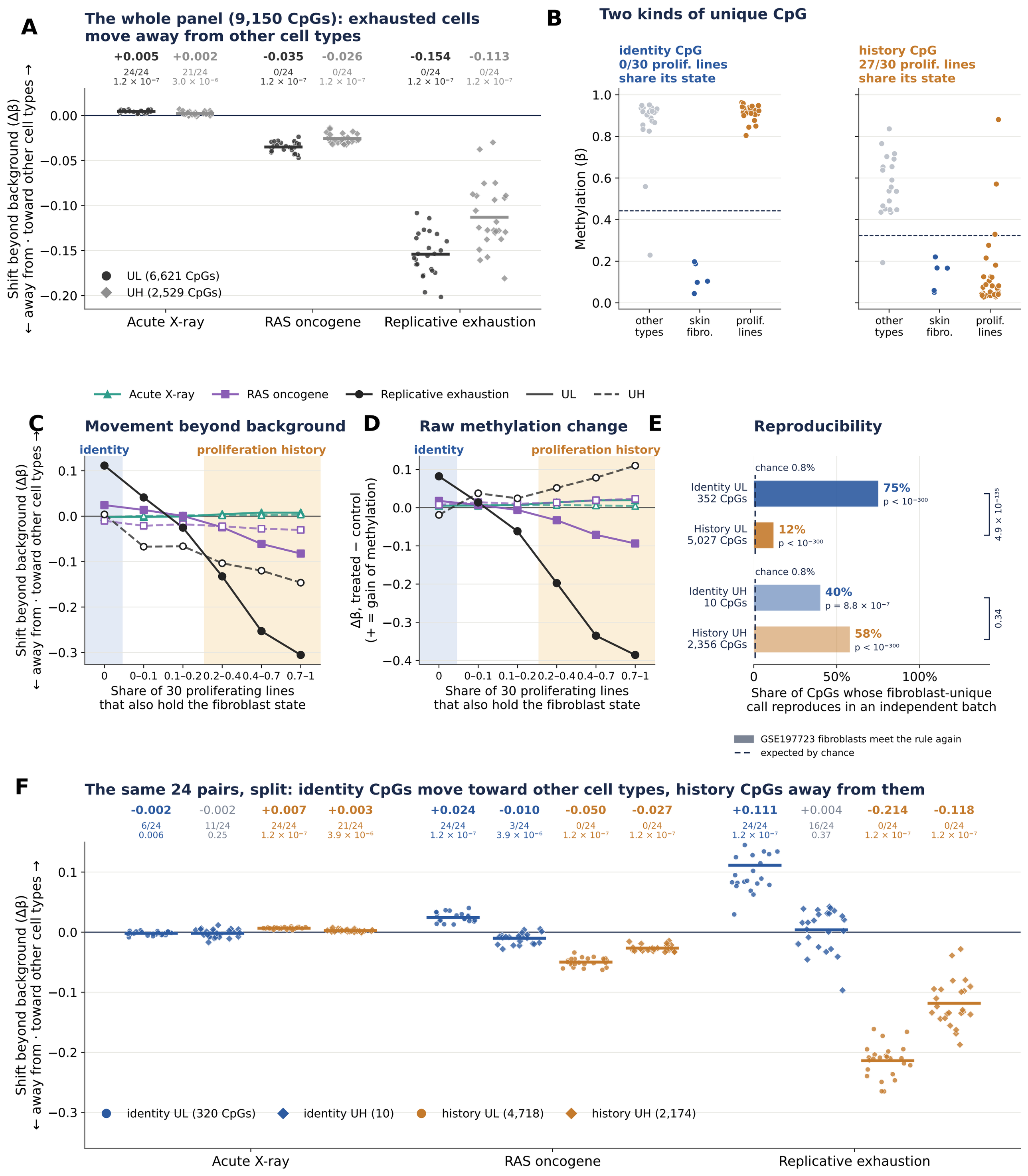

A
The whole panel (9,150 CpGs): exhausted cells move away from other cell types
+0.005
24/24
1.2 × 10⁻⁷
+0.002
21/24
3.0 × 10⁻⁶
-0.035
0/24
1.2 × 10⁻⁷
-0.026
0/24
1.2 × 10⁻⁷
-0.154
0/24
1.2 × 10⁻⁷
-0.113
0/24
1.2 × 10⁻⁷
Shift beyond background (Δβ)
← away from · toward other cell types →
0.00
−0.05
−0.10
−0.15
−0.20
UL (6,621 CpGs)
UH (2,529 CpGs)
Acute X-ray
RAS oncogene
Replicative exhaustion
B
Two kinds of unique CpG
identity CpG
0/30 prolif. lines
share its state
history CpG
27/30 prolif. lines
share its state
Methylation (β)
1.0
0.8
0.6
0.4
0.2
0.0
other types
skin fibro.
prolif. lines
Acute X-ray
RAS oncogene
Replicative exhaustion
UL
UH
C
Movement beyond background
identity
proliferation history
Shift beyond background (Δβ)
← away from · toward other cell types →
0.1
0.0
−0.1
−0.2
−0.3
0
0–0.1
0.1–0.2
0.2–0.4
0.4–0.7
0.7–1
Share of 30 proliferating lines
that also hold the fibroblast state
D
Raw methylation change
identity
proliferation history
Δβ, treated − control
(+ = gain of methylation)
0.1
0.0
−0.1
−0.2
−0.3
−0.4
0
0–0.1
0.1–0.2
0.2–0.4
0.4–0.7
0.7–1
Share of 30 proliferating lines
that also hold the fibroblast state
E
Reproducibility
chance 0.8%
Identity UL
352 CpGs
75%
p < 10⁻³⁰⁰
History UL
5,027 CpGs
12%
p < 10⁻³⁰⁰
4.9 × 10⁻¹³⁵
chance 0.8%
Identity UH
10 CpGs
40%
p = 8.8 × 10⁻⁷
History UH
2,356 CpGs
58%
p < 10⁻³⁰⁰
0.34
0
50%
100%
Share of CpGs whose fibroblast-unique
call reproduces in an independent batch
GSE197723 fibroblasts meet the rule again
expected by chance
F
The same 24 pairs, split: identity CpGs move toward other cell types, history CpGs away from them
-0.002
6/24
0.006
-0.002
11/24
0.25
+0.007
24/24
1.2 × 10⁻⁷
+0.003
21/24
3.9 × 10⁻⁶
+0.024
24/24
1.2 × 10⁻⁷
-0.010
3/24
3.9 × 10⁻⁶
-0.050
0/24
1.2 × 10⁻⁷
-0.027
0/24
1.2 × 10⁻⁷
+0.111
24/24
1.2 × 10⁻⁷
+0.004
16/24
0.37
-0.214
0/24
1.2 × 10⁻⁷
-0.118
0/24
1.2 × 10⁻⁷
Shift beyond background (Δβ)
← away from · toward other cell types →
0.1
0.0
−0.1
−0.2
−0.3
identity UL (320 CpGs)
identity UH (10)
history UL (4,718)
history UH (2,174)
Acute X-ray
RAS oncogene
Replicative exhaustion